

## EXETER, United Kingdom

WMO: 038390

Lat: 50.737N

Lon: 3.404W

Elev: 31

StdP: 100.95

Time Zone: 0.00 (EUW)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                          |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD      |              |
| (a)<br>1      | (b)<br>-4.2 | (c)<br>-2.6 | (d)<br>-5.6                   | (e)<br>2.4 | (f)<br>-2.8 | (g)<br>-4.2 | (h)<br>2.7 | (i)<br>-0.9 | (j)<br>13.3           | (k)<br>10.7 | (l)<br>12.4 | (m)<br>10.4 | (n)<br>1.3               | (o)<br>30 | (p)<br>0.579 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>9.3             | (c)<br>25.1     | (d)<br>18.2 | (e)<br>23.2 | (f)<br>17.3 | (g)<br>22.0 | (h)<br>16.9 | (i)<br>19.2         | (j)<br>23.5 | (k)<br>18.3 | (l)<br>22.0 | (m)<br>17.6 | (n)<br>20.7 | (o)<br>4.1           | (p)<br>200 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.8                            | 12.8 | 20.7 | 17.0 | 12.2 | 19.7 | 16.2 | 11.5 | 18.8 | 54.5          | 23.4 | 51.8 | 22.0 | 49.5 | 20.7 | 23.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |    |    |
|-------------------|---------|---------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|----|----|
|                   |         |         | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     | DB | WB |
| (a) 11.0          | (b) 9.6 | (c) 8.5 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |
| (a) 11.0          | (b) 9.6 | (c) 8.5 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.6   | 5.6   | 6.1   | 6.9   | 8.8   | 11.8  | 14.9  | 16.7  | 16.3  | 14.3  | 11.7  | 8.2   | 6.1  |
|                                                                                  | DBStd       | 4.96   | 3.57  | 3.08  | 2.84  | 2.65  | 2.56  | 2.30  | 2.29  | 2.24  | 2.35  | 2.89  | 3.50  | 4.24 |
|                                                                                  | HDD10.0     | 638    | 140   | 113   | 101   | 53    | 10    | 1     | 0     | 0     | 1     | 17    | 73    | 129  |
|                                                                                  | HDD18.3     | 2832   | 394   | 343   | 355   | 287   | 202   | 106   | 62    | 71    | 123   | 206   | 303   | 379  |
|                                                                                  | CDD10.0     | 868    | 4     | 3     | 4     | 16    | 67    | 147   | 206   | 194   | 129   | 69    | 20    | 9    |
|                                                                                  | CDD18.3     | 21     | 0     | 0     | 0     | 0     | 1     | 3     | 10    | 7     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 143    | 0     | 0     | 0     | 0     | 3     | 24    | 80    | 33    | 2     | 1     | 0     | 0    |
|                                                                                  | CDH26.7     | 13     | 0     | 0     | 0     | 0     | 0     | 1     | 8     | 4     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.7    | 4.3   | 4.2   | 4.2   | 4.1   | 3.6   | 3.5   | 3.5   | 3.4   | 3.4   | 3.7   | 3.7   | 3.5  |
| Precipitation                                                                    | PrecAvg     | 811    | 98    | 73    | 64    | 53    | 56    | 49    | 45    | 56    | 64    | 84    | 75    | 91   |
|                                                                                  | PrecMax     | 1285   | 224   | 163   | 132   | 143   | 148   | 119   | 111   | 129   | 186   | 311   | 210   | 195  |
|                                                                                  | PrecMin     | 524    | 11    | 2     | 0     | 3     | 3     | 1     | 14    | 10    | 13    | 7     | 18    | 14   |
|                                                                                  | PrecStd     | 146    | 52    | 46    | 29    | 37    | 35    | 31    | 25    | 28    | 45    | 65    | 47    | 44   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 14.1  | 13.2  | 16.5  | 21.0  | 23.6  | 25.9  | 27.9  | 27.2  | 23.8  | 19.3  | 16.3  | 14.4 |
|                                                                                  |             | MCWB   | 12.4  | 11.1  | 12.1  | 14.7  | 16.9  | 18.7  | 19.2  | 19.9  | 18.4  | 16.1  | 14.9  | 12.8 |
|                                                                                  | 2%          | DB     | 13.1  | 12.4  | 14.1  | 16.9  | 20.8  | 23.8  | 25.7  | 24.2  | 21.2  | 17.9  | 15.1  | 13.7 |
|                                                                                  |             | MCWB   | 11.7  | 10.5  | 11.0  | 12.7  | 15.6  | 17.4  | 18.6  | 17.9  | 17.0  | 15.5  | 13.6  | 12.5 |
|                                                                                  | 5%          | DB     | 12.1  | 11.8  | 13.0  | 15.0  | 18.8  | 22.0  | 23.8  | 22.2  | 20.0  | 17.0  | 14.2  | 12.9 |
|                                                                                  |             | MCWB   | 10.9  | 10.1  | 10.5  | 11.2  | 14.7  | 16.6  | 17.7  | 17.1  | 16.3  | 15.0  | 12.8  | 11.7 |
|                                                                                  | 10%         | DB     | 11.1  | 10.9  | 11.9  | 13.5  | 17.0  | 20.1  | 21.9  | 21.0  | 18.9  | 16.1  | 13.5  | 12.0 |
|                                                                                  |             | MCWB   | 9.9   | 9.3   | 9.8   | 10.5  | 13.6  | 15.6  | 16.9  | 16.6  | 15.6  | 14.3  | 12.1  | 10.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 13.0  | 12.0  | 12.6  | 15.0  | 17.8  | 19.6  | 20.1  | 20.6  | 19.3  | 17.1  | 15.3  | 13.3 |
|                                                                                  |             | MCDB   | 13.8  | 13.1  | 15.3  | 19.4  | 21.9  | 23.9  | 25.7  | 25.4  | 22.2  | 18.5  | 16.1  | 14.0 |
|                                                                                  | 2%          | WB     | 12.0  | 11.1  | 11.7  | 13.0  | 16.1  | 18.1  | 19.1  | 19.0  | 18.0  | 16.2  | 14.0  | 12.6 |
|                                                                                  |             | MCDB   | 13.0  | 12.3  | 13.6  | 16.2  | 19.6  | 22.0  | 24.3  | 22.4  | 20.5  | 17.4  | 14.8  | 13.5 |
|                                                                                  | 5%          | WB     | 11.2  | 10.3  | 11.0  | 11.8  | 15.1  | 17.1  | 18.3  | 18.0  | 17.2  | 15.5  | 13.2  | 11.9 |
|                                                                                  |             | MCDB   | 12.1  | 11.3  | 12.7  | 14.1  | 18.2  | 20.9  | 22.5  | 21.0  | 19.2  | 16.6  | 14.1  | 12.8 |
|                                                                                  | 10%         | WB     | 10.2  | 9.6   | 10.2  | 11.0  | 13.9  | 16.1  | 17.4  | 17.3  | 16.3  | 14.7  | 12.3  | 11.0 |
|                                                                                  |             | MCDB   | 11.0  | 10.6  | 11.6  | 13.1  | 16.5  | 19.6  | 20.9  | 20.0  | 18.2  | 15.9  | 13.4  | 12.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.4   | 6.5   | 8.0   | 8.8   | 9.1   | 9.1   | 9.3   | 9.0   | 8.8   | 7.3   | 6.7   | 6.6  |
|                                                                                  |             | MCDBR  | 6.4   | 6.4   | 9.6   | 11.3  | 11.7  | 12.4  | 12.4  | 10.8  | 10.0  | 7.4   | 5.6   | 5.0  |
|                                                                                  |             | MCWBR  | 6.1   | 5.2   | 7.0   | 7.5   | 7.4   | 7.1   | 6.4   | 5.9   | 6.4   | 5.4   | 5.0   | 4.7  |
|                                                                                  | 5% WB       | MCDBR  | 6.4   | 6.0   | 8.9   | 9.7   | 10.5  | 10.4  | 10.5  | 9.1   | 8.5   | 6.6   | 5.8   | 5.2  |
|                                                                                  |             | MCWBR  | 6.4   | 5.3   | 6.9   | 7.0   | 6.9   | 6.5   | 6.1   | 5.6   | 6.1   | 5.3   | 5.5   | 5.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.335  | 0.349 | 0.383 | 0.399 | 0.402 | 0.393 | 0.389 | 0.379 | 0.371 | 0.360 | 0.333 | 0.325 |      |
|                                                                                  | taud        | 2.339  | 2.357 | 2.266 | 2.250 | 2.279 | 2.341 | 2.357 | 2.405 | 2.415 | 2.411 | 2.393 | 2.314 |      |
|                                                                                  | Ebn at Noon | 677    | 775   | 809   | 841   | 854   | 864   | 861   | 853   | 818   | 752   | 678   | 622   |      |
|                                                                                  | Edh at Noon | 67     | 84    | 110   | 124   | 126   | 120   | 116   | 105   | 94    | 79    | 63    | 59    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.81   | 1.55  | 2.76  | 4.32  | 5.23  | 5.67  | 5.38  | 4.49  | 3.39  | 1.90  | 1.02  | 0.64  |      |
|                                                                                  | RadStd      | 0.07   | 0.17  | 0.35  | 0.40  | 0.27  | 0.57  | 0.41  | 0.36  | 0.22  | 0.14  | 0.10  | 0.08  |      |

## Historical Trends

|                              | DBAvg | Heating |     | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (57 neighbors) | N/A   | N/A     | N/A | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page